

Reasoning Reference: Clocks

Educational Series

December 29, 2025

1 Basics of a Clock

A clock is a circle of 360° divided into 12 equal divisions (hours).

- **Minute Hand:** Moves 360° in 60 minutes. Speed = $6^\circ/\text{min}$.
- **Hour Hand:** Moves 360° in 12 hours. Speed = $0.5^\circ/\text{min}$.
- **Relative Speed:** The minute hand gains 5.5° per minute over the hour hand ($6^\circ - 0.5^\circ$).

2 The Angle Formula

To find the angle θ between the hour hand (H) and the minute hand (M) at any given time:

$$\theta = \left| 30H - \frac{11}{2}M \right|$$

Note: If the angle is greater than 180° , subtract it from 360° to find the reflex angle.

3 Special Positions of Hands

In every 12-hour period:

1. **Coincidence (0°):** Hands overlap 11 times (they do not overlap between 12 and 1).
2. **Opposite (180°):** Hands are in a straight line 11 times (not between 6 and 7).
3. **Right Angle (90°):** Hands are perpendicular 22 times.

Multiply these values by 2 for a 24-hour period (e.g., they overlap 22 times in a day).

4 Finding Time for Specific Angles

To find the exact time between H and $H + 1$ hours when the hands are at an angle θ :

$$M = \frac{2}{11}(30H \pm \theta)$$

5 Faulty Clocks (Gain/Loss)

If a clock indicates 4:00 but the actual time is 3:50, the clock is **gaining** time.

- **Formula:** $\frac{\text{Incorrect Time}}{\text{Correct Time}} = \frac{\text{Incorrect Interval}}{\text{Correct Interval}}$

6 Solved Examples

Example 1: Find the angle between hands at 4:20.

- $H = 4, M = 20$
- $\theta = |30(4) - \frac{11}{2}(20)|$
- $\theta = |120 - 110| = 10^\circ$

Example 2: At what time between 7 and 8 o'clock will the hands be together?

- Together means $\theta = 0^\circ$.
- $M = \frac{2}{11}(30 \times 7 \pm 0) = \frac{2}{11}(210) = \frac{420}{11} = 38\frac{2}{11}$
- **Answer:** $38\frac{2}{11}$ minutes past 7.